Diabetes Risk Prediction Project Documentation

Introduction and Objective

Diabetes Mellitus (DM) is one of the leading chronic metabolic disorders worldwide and remains a significant public health concern. Early identification of individuals at risk enables timely intervention, helping to prevent severe complications such as cardiovascular disease, kidney failure, neuropathy, and vision loss.

The objective of this project was to develop a machine learning model capable of predicting diabetes risk using patients' clinical measurements. A Logistic Regression classifier was implemented as the primary predictive model, and its performance was evaluated before and after optimization. The project demonstrates how machine learning can support healthcare professionals by providing an efficient decision-support tool for early diabetes screening.

Dataset Description

The dataset consists of patient records containing eight clinical attributes commonly associated with diabetes risk:

- Pregnancies
- Glucose
- Blood Pressure
- Skin Thickness
- Insulin
- Body Mass Index (BMI)
- Diabetes Pedigree Function
- Age

The target variable indicates whether a patient has diabetes:

- **0** – Non-diabetic
- **1** – Diabetic

Data Cleaning

Data quality assessment revealed that several clinical variables contained zero values that are physiologically impossible and therefore represented missing observations. These variables included:

- Glucose
- Blood Pressure
- Skin Thickness
- Insulin
- BMI

Rather than removing affected records, the missing values were replaced using **median imputation**. The median was chosen because it is robust to outliers and better preserves the natural distribution of clinical measurements than the mean.

Exploratory Data Analysis (EDA)

Several visualizations were created to understand the dataset before model development.

Class Distribution

The count plot showed that approximately **65%** of the patients were non-diabetic, while **35%** were diabetic. Although the imbalance was moderate, it justified later experimentation with SMOTE during model optimization.

Glucose Distribution

A box plot comparing glucose levels by diabetes outcome revealed that diabetic patients consistently exhibited higher glucose concentrations than non-diabetic patients. Glucose showed the clearest separation between the two classes.

BMI and Glucose Relationship

A scatter plot demonstrated that patients with both elevated BMI and high glucose levels were considerably more likely to be diabetic, highlighting the combined influence of obesity and blood sugar on diabetes risk.

Age Distribution

The age distribution showed that while younger adults formed the majority of the dataset, the proportion of diabetic patients increased steadily with age.

Feature Correlation

The correlation heatmap indicated that **Glucose** had the strongest positive relationship with diabetes outcome, followed by **BMI** and **Age**, confirming their importance as predictive variables.

Data Splitting

To evaluate model performance objectively, the cleaned dataset was divided into:

- **80% Training Set (614 observations)**
- **20% Testing Set (154 observations)**

The testing data remained completely unseen during model training and optimization to ensure an unbiased evaluation.

Feature Scaling

Since Logistic Regression is sensitive to differences in feature scales, all predictor variables were standardized using **StandardScaler**.

Standardization transformed each feature to have:

- Mean = 0
- Standard Deviation = 1

This ensured that variables measured on different numerical scales contributed equally during model training.

Baseline Logistic Regression Model

A baseline Logistic Regression model was first trained using the standardized training dataset without any additional optimization.

The purpose of this model was to establish a benchmark for evaluating whether optimization techniques could improve predictive performance.

Addressing Class Imbalance Using SMOTE

Although the dataset exhibited only moderate class imbalance, machine learning models often become biased toward the majority class.

To improve the model's ability to recognize diabetic patients, the **Synthetic Minority Oversampling Technique (SMOTE)** was incorporated into the optimized training pipeline.

SMOTE creates synthetic minority class samples rather than duplicating existing observations. This enables the classifier to learn more representative decision boundaries while reducing bias toward the majority class.

Importantly, SMOTE was applied **only to the training data** through the machine learning pipeline, preventing information leakage into the test dataset.

Hyperparameter Optimization

To improve model performance, **GridSearchCV** with **5-fold cross-validation** was used to identify the optimal Logistic Regression hyperparameters.

The optimization process evaluated different combinations of:

- Regularization strength (C)
- Solver algorithm
- Penalty type

The grid search was configured to maximize **Recall**, since correctly identifying diabetic patients is more clinically important than maximizing overall accuracy.

Machine Learning Pipeline

To ensure a robust workflow, the optimized model was implemented using a machine learning pipeline consisting of:

1. StandardScaler
2. SMOTE
3. Logistic Regression

Using a pipeline ensured that scaling and oversampling were performed correctly within each cross-validation fold, preventing data leakage and producing reliable model evaluation.

Feature Importance

Following model training, the Logistic Regression coefficients were examined to identify the most influential predictors.

The analysis showed that:

- **Glucose** contributed the most toward predicting diabetes.
- **BMI** was the second strongest predictor.
- **Age** also showed a substantial positive influence.

These findings align with established medical knowledge regarding diabetes risk factors.

Model Evaluation

Both models were evaluated using the same unseen testing dataset.

Baseline Logistic Regression

The baseline model achieved strong overall performance and accurately classified most patients. However, approximately **31%** of diabetic patients were incorrectly classified as non-diabetic.

Optimized Logistic Regression (StandardScaler + SMOTE + GridSearchCV)

The optimized model significantly improved recall, successfully identifying a greater proportion of diabetic patients while maintaining comparable overall performance.

Model Comparison

Metric	**Baseline Model**	**Optimized Model**
Accuracy	**82%**	80%
Precision	**77%**	69%
Recall	69%	**76%**
F1-score	73%	73%

The baseline model produced the highest overall accuracy and precision, indicating fewer false-positive predictions.

The optimized model, however, substantially improved recall, increasing the proportion of diabetic patients correctly identified from **69% to 76%**. Although this resulted in a slight reduction in accuracy and precision, the optimized model maintained the same F1-score, demonstrating a balanced trade-off between sensitivity and overall performance.

For healthcare screening applications, improving recall is particularly valuable because failing to identify a patient with diabetes may delay diagnosis and treatment.

Confusion Matrix Interpretation

Baseline Logistic Regression

The confusion matrix for the baseline model showed the following results:

- **True Negatives (TN): 89** – Healthy individuals correctly classified as non-diabetic.
- **False Positives (FP): 11** – Healthy individuals incorrectly classified as diabetic.
- **False Negatives (FN): 17 –** Diabetic patients incorrectly classified as healthy.
- **True Positives (TP): 37** – Diabetic patients correctly identified by the model.

The model correctly identified the majority of healthy patients while detecting **37 out of 54 diabetic patients**. However, it failed to identify **17 diabetic patients**, contributing to a recall of **69%**. Although the baseline model achieved high overall accuracy, the relatively high number of false negatives indicates that some diabetic cases would go undetected if the model were used for screening.

Optimized Logistic Regression (StandardScaler + SMOTE + GridSearchCV)

The confusion matrix for the optimized model produced the following results:

- **True Negatives (TN): 82** – Healthy individuals correctly classified as non-diabetic.
- **False Positives (FP): 18** – Healthy individuals incorrectly classified as diabetic.
- **False Negatives (FN): 13 –** Diabetic patients incorrectly classified as healthy.
- **True Positives (TP): 41 –** Diabetic patients correctly identified by the model.

Compared with the baseline model, the optimized model correctly identified **41 diabetic patients**, an increase of four additional positive cases. At the same time, the number of false negatives decreased from **17 to 13**, improving the model's recall from **69% to 76%**.

This improvement came at the expense of an increase in false positives, which rose from **11 to 18**. Consequently, while more healthy individuals were incorrectly flagged as diabetic, fewer diabetic patients were missed. In a healthcare screening context, this trade-off is generally acceptable because patients identified as high risk can undergo additional diagnostic testing, whereas missed diabetic cases may delay treatment and increase the risk of complications.

Confusion Matrix Comparison

Metric	Baseline	Optimized
True Negatives	**89**	82
False Positives	**11**	18
False Negatives	17	**13**
True Positives	37	**41**

The comparison demonstrates the effect of applying SMOTE and hyperparameter optimization. The optimized model became more effective at detecting diabetic patients, correctly identifying four additional positive cases and reducing missed diagnoses by four patients. Although this improvement increased the number of false alarms among healthy individuals, the reduction in false negatives makes the optimized model more appropriate for diabetes screening, where identifying as many true diabetic patients as possible is the primary objective.

Conclusion

This project successfully developed and evaluated two Logistic Regression models for diabetes risk prediction.

The baseline model provided strong overall classification performance, achieving an accuracy of **82%** and a precision of **77%**.

The optimized model incorporated feature scaling, SMOTE, and GridSearchCV, improving recall from **69%** to **76%** while maintaining an F1-score of **73%** and an overall accuracy of **80%**.

The results demonstrate that optimization techniques can improve the model's ability to identify diabetic patients while maintaining competitive predictive performance.

Recommendation

While the baseline model achieved slightly higher overall accuracy, the **Optimized Logistic Regression model is recommended for diabetes risk screening**.

The improved recall means that more diabetic patients are correctly identified, reducing the number of missed positive cases. In healthcare, false negatives often have more serious consequences than false positives because delayed diagnosis can postpone treatment and increase the risk of complications.

Therefore, the optimized model is better suited as a **clinical decision-support tool** for early diabetes screening rather than as a standalone diagnostic system.

Future work could further improve performance by:

- Incorporating additional clinical features such as HbA1c, family history, and lifestyle factors.
- Exploring ensemble learning algorithms such as Random Forest, XGBoost, or LightGBM.
- Performing probability threshold tuning to further optimize the balance between precision and recall.
- Validating the model on larger and more diverse patient populations before clinical deployment.